Earthquakes

1. Give two reasons why most earthquakes are not felt.

2. Explain why friction exists at plate boundaries.

3. Use friction to explain what causes earthquakes.

4. Explain the difference between the focus and the epicentre of an earthquake.

5. What is the maximum depth the focus can be below the surface? Explain why it can be no deeper.

6. Seismic waves can be split into two main categories, body waves and surface waves.

    a. Body Waves are

    b. Surface Waves are

7. A tremor is an earthquake that can be felt but does little damage. What would be its value on both the Richter and Mercalli scales?

8. At what value on the Richter scale would you call and earthquake ‘serious’? Explain why you chose this number.

9. Aftershocks are often more dangerous than the first earthquake. Explain why this is the case.

10. Why is fire a big risk after an earthquake?

11. Explain what causes a tsunami to form and why they are so dangerous to people living on coastlines.

12. Australia has few earthquakes, yet our neighbours Papua New Guinea and New Zealand have lots. Use the theory of plate tectonics to explain why.

13. Some of the statements below are incorrect. Modify the incorrect statements so they become true.

    a. Aftershocks are huge waves caused by earthquakes.

    b. An earthquake of strength 5 on the Richter scale is double the strength of an earthquake of strength 4.

    c. Earthquakes are caused by plates slipping.

    d. The focus of an earthquake is the exact point where it starts.

    e. Tsunamis are largest when in deep water.